Huawei Access Node HSR-ACC-74: Standard Operating Procedure (SOP) for Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) Failure

Node: HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4537

Description

This SOP outlines the steps to investigate and resolve a major failure on Huawei Access Node HSR-ACC-74, located at HSR Layout Access, due to a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding). The failure is characterized by the alarm code ALARM_4537. The protocol mechanics behind this failure involve the dynamic convergence of routing protocols, which can cause a flap in the routing table, leading to packet loss and service disruption. In a Huawei Access Node, the BGP, OSPF, and ISIS routing protocols interact with each other, and a malfunction in any one of these protocols can cause a flap in the routing table. The Huawei Access Node's role in this scenario is critical, as it acts as the point of convergence for multiple networks, and its failure can have a significant impact on the overall network availability. To mitigate this failure, it is essential to identify the root cause of the flap and take corrective action to stabilize the routing protocols.

Pre-requisite Checks

- Verify that the HSR-ACC-74 node is accessible via the Huawei CLI.
- Check the redundant link status to ensure that it is functioning correctly.
- Dispatch remote hands to HSR Layout Access to perform any necessary physical checks.

Potential Root Causes

Cause	Probability
BGP route churn due to neighbor peer down	High
OSPF adjacency loss due to neighbor peer down	Medium
ISIS LSP flooding due to neighbor peer down	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp peer, Expected Output: BGP peer information: ..., Description: This command displays the BGP peer information, including the neighbor IP address, state, and timers. It can help identify any issues with the BGP neighbor peer connection.

Command: display ospf neighbor, Expected Output: OSPF neighbor information: ..., Description: This command displays the OSPF neighbor information, including the neighbor IP address, state, and timers. It
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can help identify any issues with the OSPF neighbor peer connection.

Command: display isis lsp, Expected Output: ISIS LSP information: ..., Description: This command displays the ISIS LSP information, including the LSP ID, state, and timers. It can help identify any issues with the ISIS LSP flooding.
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Resolution Steps

Step 1: Step: Verify BGP neighbor peer connection, Command: display bgp peer, Description: Verify that the BGP neighbor peer connection is stable and functioning correctly.

Step 2: Step: Reset BGP neighbor peer connection, Command: reset bgp peer , Description: Reset the BGP neighbor peer connection to re-establish communication with the neighbor peer.

Step 3: Step: Verify OSPF neighbor peer connection, Command: display ospf neighbor, Description: Verify that the OSPF neighbor peer connection is stable and functioning correctly.

Step 4: Step: Reset OSPF neighbor peer connection, Command: reset ospf neighbor , Description: Reset the OSPF neighbor peer connection to re-establish communication with the neighbor peer.

Step 5: Step: Verify ISIS LSP flooding, Command: display isis lsp, Description: Verify that the ISIS LSP flooding is stable and functioning correctly.

Step 6: Step: Reset ISIS LSP flooding, Command: reset isis lsp, Description: Reset the ISIS LSP flooding to re-establish communication with the neighbor peer.

Post-Check Verification

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Command: display bgp peer, Expected Output: BGP peer information: ..., Description: Verify that the BGP peer information is stable and functioning correctly.

Command: display ospf neighbor, Expected Output: OSPF neighbor information: ..., Description: Verify that the OSPF neighbor information is stable and functioning correctly.

Command: display isis lsp, Expected Output: ISIS LSP information: ..., Description: Verify that the ISIS LSP information is stable and functioning correctly.
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Roll-back Procedures

- Step: Restore BGP neighbor peer connection, Command: undo reset bgp peer , Description: Restore the BGP neighbor peer connection to its previous state.

- Step: Restore OSPF neighbor peer connection, Command: undo reset ospf neighbor , Description: Restore the OSPF neighbor peer connection to its previous state.

- Step: Restore ISIS LSP flooding, Command: undo reset isis lsp, Description: Restore the ISIS LSP flooding to its previous state.